

tion to add an offset resistor between pin 12 and GND. The offset resistor has to be larger than the frequency-selection resistor R1, but otherwise can be infinite.

All we need next is a simple variable voltage source to test it. For that, the 100k linear potentiometer (VR1) can be hooked up, as shown in the schematic diagram. And then, VR1 can be used to vary the output frequency from around 0Hz to 1kHz at ease. Fig. 4 shows the frequency output at pin 4, which can be seen on an oscilloscope.

A few more key things worth noting are:

- VCO output across CON3 is the output point of the VFC module.

- CON1 is the 0-5V DC input connector of the VFC module.

- VR1 is an optional component, but it is helpful to test the VFC module. However, after construction and initial testing, it should be removed from the setup.

- LED1 (red/amber) is included deliberately to provide a visual indication of the output signal. The NXP datasheet does not include specific ratings for the VCO output, but seems to cater nothing greater than ~10mA. Since HEF4046B does not really appear to be tailored to drive any significant load, LED1 is also an optional component.

- Since the voltage input (at pin 9) of IC1 is very sensitive, if it's not connected to anything, it might pick up power-line interferences and change the frequency wildly. Adding an appropriate capacitor (C2) can solve this problem, whether it's useful or not depends on you.

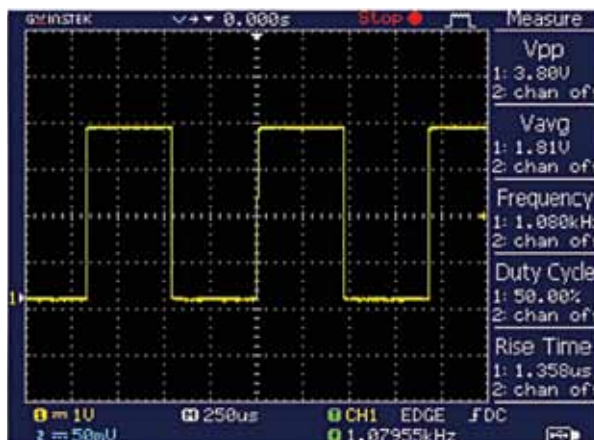


Fig. 4: Frequency output at pin 4 shown on oscilloscope

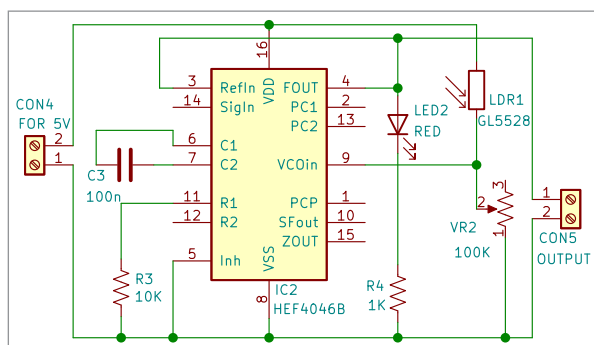


Fig. 5: VFC module circuit using LDR

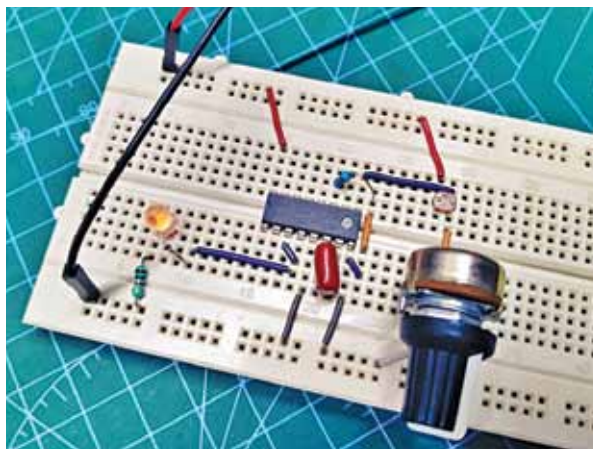


Fig. 6: VFC module using LDR wired on a breadboard

Analogue sensor test using LDR (optional)

Now, let us do something that could not be done with a potentiometer as the frequency control element, that is, control the VCO frequency through an analogue sensor! To do that, follow the schematic diagram

shown in Fig. 5, where a common GL5528 LDR (<https://pi.gate.ac.uk/pages/airpi-files/PD0001.pdf>) is connected to the voltage input of IC1 as the analogue sensor.

Here, the photoresistor (LDR1) and the potentiometer (VR2) combine to form a potential divider, so that the output frequency varies from 0Hz (completely dark) to 1kHz (strong light), depending on the intensity of the light falling on the photoresistor (see the oscillogram below). Another type of analogue sensor that can be used here is a thermistor; the frequency then will depend on the temperature.

Nevertheless, the analogue sensor setup is just about the same functionality as the potentiometer-tuned voltage controlled oscillator from before. However, VR2 does not play a crucial role at this time, so it may be substituted with a 100k fixed resistor. (It's pretty easy if you know what you're doing.) Fig. 6 shows the VFC module with LDR (optional) wired on a breadboard.

EFY note. This is just a design concept for a crude, hobby-level VFC module. In fact, a real VFC circuit is somewhat different than this. Keep in mind that VFC is a special type of VCO designed to be very linear over a wide range of input voltages, and it features many advantages and enhanced performance

specifications, which a VCO doesn't have. There are better designs out there, but they all probably demand dedicated ICs like AD654. **EFY**



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